MIL-F-60083E(EA)
23 April 1974
SUPERSEDING
MIL-F-60083D(MU)
1 October 1970

MILITARY SPECIFICATION

FUZE, BOMB, ELECTRIC, FMU-7C/B

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of fuze.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-781	-	Strapping, Steel, Flat and Seals.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Box, Fiberboard.
PPP-F-320	-	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

STANDARDS

FEDERAL

Fed Std No. 595	-	Colors.

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-331	-	Fuze and Fuze Components, Environmental and Performance Tests for.
MIL-STD-810	-	Environmental Test Methods.
MIL-STD-1168	-	Lot Numbering of Ammunition.

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DRAWINGS

US ARMY ARMAMENT COMMAND

EDGEWOOD ARSENAL

E14-15-913 - Fuze, Bomb, Electric, FMU-7C/B.
IDL14-15-913 - Fuze, Bomb, Electric, FMU-7C/B - Inspection Data List.

PUBLICATIONS

EDGEWOOD ARSENAL INSPECTION STANDARDS

R7-6 - Radiographic Evaluation of Fuze, Bomb, Electric, FMU-7C/B.
R14-13 - Soundness and Surface Condition of Zinc Die Cast Components for Fuze AN-M173A1 and Fuze FMU-7C/B.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing E14-15-913 or on the subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the fuze shall conform to the specifications and drawings listed on Drawing E14-15-913 and on subsidiary drawings.

3.2 Assembly. The fuze shall be assembled as shown on Drawing E14-15-913 (see 6.3).

3.3 Striker assembly (Drawing C14-15-909). The striker assembly shall comply with the following when tested as specified in 4.4.4.1.

(a) A 0.095-inch diameter steel pin gage with a hemispherical end shall not pass between the balls.

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(b) A 0.087-inch diameter steel pin gage with a hemispherical end shall pass between the balls freely.

(c) The striker assembly, with the 0.087 inch diameter steel pin in place between the balls, shall be inserted into a sleeve (Dwg C14-15-1006). The striker assembly shall drop of its own weight into the sleeve upon removal of the steel pin. NOTE: For safety, the sleeve used during the test shall not have a primer present.

3.4 Safety. The fuze less the detonator and booster cup shall not function when dropped primer side down from a height of 7 feet ± 1 inch when tested as specified in 4.4.4.2.

3.5 Continuity. Subsequent to the safety test, the assembled fuze shall possess a resistance of 2 to 9 ohms when tested as specified in 4.4.4.3.

3.6 Arming and functioning. After conditioning in accordance with 4.4.4.8 the fuze shall comply with the following:

3.6.1 Arming. When tested as specified in 4.4.4.4(a), the fuze with primer and detonator, but without tetryl pellets shall arm between 5 and 30 milliseconds under an input of 1.3 volts DC supplied by a constant voltage source. The arming time is the interval, measured to the nearest 0.10 millisecond, between the closing of the electrical circuit and the piercing of the foil by the arming pin.

3.6.2 Functioning. The armed fuze complete with primer and detonator, but without tetryl pellets shall function upon impact when dropped with random orientation from a height of 12 ± 1 foot (see 6.3) and tested as specified in 4.4.4.4(b).

3.7 Jolt. The assembled fuze without detonator and tetryl shall comply with the requirements of MIL-STD-331 when tested as specified in 4.4.4.5.

3.8 Jumble. The assembled fuze without detonator and tetryl shall comply with the requirements of MIL-STD-331 when tested as specified in 4.4.4.6.

3.9 Forty-foot drop. The assembled fuze without detonator and tetryl when dropped at random orientation shall comply with the requirements of MIL-STD-331 when tested as specified in 4.4.4.7.

3.10 Vibration. The assembled fuze without tetryl (see 6.3) shall be conditioned as specified in 4.4.4.8 before testing for compliance with 3.6.

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3.11 Radiographic examination. The fuze less the detonator and booster cup shall be examined radiographically in accordance with 4.4.4.9.

3.12 Soundness and surface conditions. The strikerbody, case, booster cup, cup retainer, body, sleeve and cap shall be examined in accordance with 4.4.4.10 after completion of all machining operations.

3.13 Torque. There shall be no movement between the body and the case and the booster cup when a breakaway torque of 40 inch-pounds is applied when tested in accordance with 4.4.4.11.

3.14 Preproduction. Prior to the start of regular production, a preproduction lot of fuzes shall be produced in accordance with this specification for examination and test (see 4.3).

3.15 Workmanship. The fuze shall be free from foreign matter (oil and grease) and damage such as dents, cracks or deformation which could affect its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contrating officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections.

(a) Preproduction inspection (see 4.3)

(b) Quality conformance inspection (see 4.4)

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4.3 Preproduction inspection:

4.3.1 Lot. A preproduction lot of 250 fuzes shall be manufactured using the same methods, materials, equipment and processes as will be used during regular production.

4.3.2 Sampling.

4.3.2.1 For examination and nondestructive tests. Each fuze in the preproduction lot shall be taken for critical defects. Each fuze in the preproduction lot shall be taken for continuity test. Each striker assembly in the preproduction lot shall be taken for test. Fifty fuzes shall be taken at random from the preproduction lot.

4.3.2.2 For destructive tests. Eighty fuzes shall be taken at random from the preproduction lot.

4.3.3 Inspection procedure:

4.3.3.1 For examination and nondestructive tests. Each fuze in the preproduction lot shall be tested in accordance with 4.4.4.2, 4.4.4.3 and 4.4.4.9. Each striker assembly in the preproduction lot shall be tested in accordance with 4.4.4.1. The sample fuzes shall be examined and tested in accordance with the classification of defects (4.4.3.4). The AQL's do not apply.

4.3.3.2 For destructive tests. The sample fuzes shall be tested in accordance with table I.

Table I. Preproduction sampling and testing

Quantity	Requirement	Test
	3.10	4.4.4.8
*50	3.6	4.4.4.4
10	3.7	4.4.4.5
10	3.8	4.4.4.6
10	3.9	4.4.4.7

*Condition fuzes in accordance with 4.4.4.8; then test in accordance with 4.4.4.4.

4.3.4 Acceptance/rejection criteria. The preproduction sample shall comply with the examinations and tests specified in 4.3.3 to be acceptable. The acceptance number is zero. The supplier shall obtain written approval from the contracting officer prior to proceeding with regular production.

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4.4 Quality conformance inspection.

4.4.1 Lotting. Each lot shall be identified and controlled in accordance with MIL-STD-1168. However, no more than interfix number of each component (detonators, pellets, primers and motor assembly bellows) from any one producer shall be contained in any one lot of assembled fuzes.

4.4.2 Sampling.

4.4.2.1 For examination and nondestructive tests. Sampling shall be conducted as follows:

(a) Classification of defects. In accordance with MIL-STD-105.

(b) Striker assembly. Each striker assembly in the lot shall be taken for test.

(c) Continuity. Each fuze in the lot shall be taken for test.

4.4.2.2 For destructive tests. Sampling shall be conducted as follows:

(a) Arming and functioning. In accordance with MIL-STD-105, level S-4.

(b) Jolt, jumble and 40-foot drop. In accordance with MIL-STD-105, level S-2 for each test.

4.4.3 Inspection procedure.

4.4.3.1 For examination and nondestructive tests. The sample fuzes and striker assemblies shall be examined and tested as follows:

(a) Classification of defects. In accordance with the classification of defects and with MIL-STD-105.

(b) Striker assembly. In accordance with 4.4.4.1 and 4.4.5.

(c) Continuity. In accordance with 4.4.4.3 and 4.4.5

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4.4.3.2 For destructive tests. The sample fuzes shall be tested as follows:

(a) Arming and functioning. After vibration (4.4.4.8), test in accordance with 4.4.4.4 and MIL-STD-105 using an AQL of 1.5 percent defective.

(b) Jolt, jumble and 40 foot drop. In accordance with 4.4.4.5, 4.4.4.6, 4.4.4.7 and MIL-STD-105 using an AQL of 1.5 percent defective.

4.4.3.3 For critical defects. Each fuze in the lot shall be tested for its critical characteristics listed in the classification of defects.

4.4.3.4 Classification of defects. (See inspection data list IDL14-15-913 for inspection equipment required to perform the examinations specified in this section.)

(a) Fuze, bomb, electric, FMU-7C/B (Drawing E14-15-913).

Categories	Defects	Acceptance standards
Critical:		
1	Fuze safety	4.4.4.2
2	Radiographic examination	4.4.4.9
Major:	AQL 1.0 percent defective	
101	Thread (1-1/2-24NS-2A) incorrect	
102	Component missing or incorrectly assembled	
103	Marking missing or incorrect	
104	Workmanship (3.15)	
105	Torque incorrect	4.4.4.11
106	Arming pin not visible	

(b) Sleeve assembly (Drawing C14-15-907).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Primer inverted

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4.4.3.4 Classification of defects (continued).

(c) Striker assembly (Drawing C14-15-909).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Striker point profile incorrect
102	Length (0.684) incorrect

(d) Striker spring (Drawing C14-15-991).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Striker spring load at 0.156 incorrect

(e) Striker body (Dwg C14-15-188), Case (Dwg C14-15-277), Cup, booster (Dwg C14-15-278), Retainer cup (Dwg B14-15-282), Body (Dwg E14-15-910), Sleeve (Dwg C14-15-1006) and Cap (Dwg C14-15-905).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 1.0 percent defective	
101	Soundness and surface condition incorrect	4.4.4.10

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4.4.3.4 Classification of defects (continued).

(f) Preparation for delivery (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Container incorrect
102	Container incorrectly closed
103	Marking missing, incorrect or illegible
104	Strapping missing

4.4.4 Tests. Tests shall be conducted as follows: (See inspection data list IDL14-15-913 for inspection equipment required to perform the tests specified in this section.)

4.4.4.1 Striker assembly. The assembly shall be checked for adequate staking by inserting first a 0.087 inch diameter pin gage and second a 0.095 inch diameter pin gage into the 0.097 inch diameter hole. With the 0.087 inch diameter pin gage in place between the balls, the striker assembly shall be inserted into a sleeve (Dwg C14-15-1006). With the sleeve in an upright position, the 0.087 inch diameter pin shall be withdrawn from the striker assembly.

4.4.4.2 Safety. The fuze assembly without the M31A1 detonator and booster cup with tetryl pellets shall be dropped, primer down through a tube of suitable diameter from the specified height onto a rubber slab. The rubber slab shall be 1/4 + 1/8 inch thick, possess a durometer reading of 40 ± 10 and be supported on the entire lower surface area by a 1 inch thick steel plate.

4.4.4.3 Continuity. The fuze assembly continuity shall be determined using a continuity tester incapable of passing a current of more than 10 milliamperes or an Alinco Tester No. 101-5BP6.

4.4.4.4 Arming and functioning. The fuze complete with primer, detonator and empty booster cup (without tetryl pellets) shall be tested as follows:

(a) Arming.

(1) Connect the fuze to the power source.

(2) Connect the fuze to the recording device capable of measuring the arming time.

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(3) Close the electrical circuit and record the arming time.

(b) Functioning. Drop the fuze from the required height to impact on a firmly supported steel plate or concrete slab. The fuze shall be released in a random manner so that it will have an equal chance of landing at any angle upon impact.

4.4.4.5 Jolt. The fuze shall be tested in accordance with test 101 of MIL-STD-331.

4.4.4.6 Jumble. The fuze shall be tested in accordance with test 102 of MIL-STD-331.

4.4.4.7 Forty-foot drop. The fuze shall be tested in accordance with test 103 of MIL-STD-331 without the bomb.

4.4.4.8 Vibration. The fuze shall be conditioned in accordance with Method 514.1, table 514.1 - VII, equipment category g (shipment by common carrier), procedure X and curve AW of MIL-STD-810.

4.4.4.9 Radiographic examination. Each fuze less detonator and booster cup shall be examined and evaluated in accordance with Edgewood Arsenal Inspection Standard R7-6.

4.4.4.10 Soundness and surface conditions. The components shall be examined and evaluated in accordance with Edgewood Arsenal inspection Standard R14-13.

4.4.4.11 Torque. The fuze shall be placed in a fixture and the specified load applied against the fuze case and booster cup.

4.4.5 Acceptance/rejection criteria. If any striker assembly fails the test in 4.4.4.1, that striker assembly shall be rejected. If any fuze fails the continuity test 4.4.4.3, that fuze shall be rejected. Items passing the above tests shall be returned to their lots.

5. PREPARATION FOR DELIVERY

5.1 Packaging, level A. The fuzes shall be packaged, two (2) per container, in accordance with Drawing C14-15-1023.

5.2 Packing, level shall be as specified (see 6.2).

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5.2.1 Level A. The fuzes packaged in accordance with 5.1, shall be packed 14 cans (28 fuzes) per box, in a wood box conforming to Class 2, style 4, Grade A, 100-250 pound load limit for type 1 or 2 load for overseas shipment, PPP-B-621, having inside dimensions in inches of 17-1/2 by 9-3/4 by 5-7/8 length, width and depth respectively. Arrangement shall be two rows of seven cans each. Filler pads of weather-resistant fiberboard conforming to PPP-F-320, shall be included at each interior face of the box, with additional pads as necessary to restrict free movement of the cans. Boxes shall be closed and strapped in accordance with the appendix to PPP-B-621. Strapping shall conform to Class A or B of QQ-S-781.

5.2.2 Level B. Packing shall be in accordance with 5.2.1, except the box shall be Grade B, non-preservative treated.

5.2.3 Level C. The fuzes shall be packaged in accordance with 5.1, and packed, 14 cans (28 fuzes) per box, in fiberboard boxes having inside dimensions in inches of 17-1/4 by 9-1/2 by 5-3/4 length, width, and depth, respectively and conforming to style RSC, Grade V3c, PPP-B-636. Arrangement shall be two rows of seven cans per row. The boxes shall be closed by gluing in accordance with Method III in the appendix to PPP-B-636 and reinforced by banding with tape or non-metallic strapping in accordance with said appendix.

5.3 Palletization. When specified in the contract or order (see 6.2), fuzes packed as specified in 5.2, shall be palletized in accordance with Drawing D14-15-997.

5.4 Marking. In conjuction with information provided in the contract or order (see 6.2) marking shall be as follows:

5.4.1 Package. Each can of fuzes shall be marked with the lot number, federal stock number, Department of Defense ammunition code number, nomenclature, quantity and unit of issue, contract or purchase order number, level of packaging and month and year packaged. In addition the can shall be provided with a 1/2 inch wide band of color yellow, number 23538 or 33538 of FED-STD-595, with the word "Tetryl" in 1/4 inch high letters applied on band on two opposite sides. The band shall be located around the circumference of the can in a manner not to interfere with other required marking.

5.4.2 Packing. Boxes shall be marked in accordance with Drawing D14-15-988.

5.4.3 Palletized loads. When fuzes are palletized, pallets shall be marked in accordance with Drawing D14-15-997.

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6. NOTES

6.1 Intended use: This item is intended to initiate fire bombs BLU-1B/B, BLU-10/B, BLU-23/B, BLU-27/B and BLU-32.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Palletization if required (mandatory for overseas shipment).
(c) Level of packing required.
(d) Pertinent identification marking.
(e) Applicable address marking.
(f) When wood preservative is required for pallets (mandatory for overseas shipment).
(g) Shipment priority number, if applicable:
(h) Transportation control number, project code and required delivery date, if applicable.
(i) Any special marking required:
(j) Preproduction

(1) Time allowed for supplier's submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the supplier whether to proceed with production.

6.3 Caution. Suppliers are hereby notified to take all necessary precautions to protect personnel to prevent explosions while preparing the mixture, assembly and testing the fuze. The supplier may, at his own discretion, omit the detonator during vibration or arming.

6.4 Component specifications. The following specifications reflect the components required for the complete assembly of FMU-7C/B fuze.

MIL-B-55291 - Battery, Thermal.
MIL-M-60079 - Motor Assembly, Bellows.
MIL-I-60084 - Initiator, Bomb Fuze, FMU-7C/B.
MIL-C-60085 - Cable Assembly, FMU-7C/B.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 1325-A452

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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